

Design, Modeling, and Control of a PVC-Based Electric Go-Kart with Embedded Traction Control and VRLA Battery Management System

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GitHub: <https://github.com/aadityakamble18/Electric-GoKart>

Abstract—This paper presents an in-depth engineering study of a low-cost electric go-kart incorporating a PVC-based chassis, dual BLDC hub motors, an Arduino-based traction control system (TCS), and a custom VRLA battery management system (BMS). The system is analysed from a multi-domain perspective, including mechanics, electrical systems, control theory, and embedded implementation. MATLAB simulations and GUI-based visualisation tools are used to validate the dynamic behaviour of the system. The work emphasises physical intuition, mathematical modelling, and real-time control strategies to ensure stable and safe vehicle operation. Simulation results demonstrate that the TCS corrects wheel-slip events within 200–250 ms and that the BMS successfully enforces all three protection thresholds (undervoltage, overcurrent, overtemperature) without nuisance trips during a 300 s drive cycle.

I. INTRODUCTION

The transition towards electric mobility has necessitated the development of efficient, safe, and intelligent energy systems. While commercial electric vehicles (EVs) incorporate advanced control systems, these systems are often inaccessible in academic or low-cost prototyping environments [1]. This project bridges that gap by developing a complete electric vehicle platform using accessible materials and open-source electronics.

The novelty of this work lies not only in the physical construction of the go-kart but in the deep integration of control systems and battery safety mechanisms. The system is designed to mimic real-world EV architectures at a smaller scale, allowing detailed study of motor dynamics, traction behaviour, and battery characteristics [2].

A key challenge addressed in this work is the absence of a mechanical differential. Dual hub motors inherently behave like an open differential, where torque distribution is not regulated, leading to instability. This necessitates the implementation of an electronic traction control system [3]. Traction control systems (TCS) are well-established in commercial vehicles and typically rely on wheel-speed sensing to detect and correct slip [4]. Adapting such systems to a low-cost Arduino platform represents the core embedded contribution of this work.

The go-kart integrates a PVC chassis, two 36 V 350 W BLDC hub motors, two independent motor controllers, an Arduino-based TCS, a VRLA Battery Management System, and a precharge-and-contactor safety circuit. Software tools used include MATLAB, Arduino IDE, SolidWorks, Ansys, and basic electrical simulation environments. All simulation source files are available at the project repository [5].

II. LITERATURE REVIEW

A. PVC-Based Structural Prototypes

PVC (polyvinyl chloride) has been studied as a structural material in low-load applications. Research on PVC pipe trusses and student robotics platforms confirms that Schedule-40 PVC can withstand moderate static and quasi-dynamic loads when joints are made with solvent-cement bonding, which creates chemically fused, monolithic connections [6]. Compared to welded steel, PVC offers ease of machining, rapid assembly, and substantial cost savings, making it well-suited for academic EV prototypes [7].

B. VRLA Batteries in Low-Power EVs

Valve Regulated Lead-Acid (VRLA) batteries are widely deployed in mobility scooters, golf carts, and UPS systems due to their robustness, sealed maintenance-free operation, and ability to deliver high surge currents [8]. Key drawbacks include lower energy density relative to lithium-ion chemistries and significant voltage sag under high current loads, characterised by the Peukert effect [9]. Monitoring and protection strategies for VRLA packs must therefore account for both electrochemical nonlinearity and thermal degradation [10].

C. Brushless DC Motor Control

BLDC hub motors with integrated hall-effect sensors are standard in e-bikes and light electric vehicles. Off-the-shelf controllers implement trapezoidal commutation and accept analogue throttle inputs but universally lack native traction control capability [11]. Externally imposed traction control via a microcontroller interposed between the throttle and the controller is an under-explored approach at the hobby and academic level [12].

D. Traction Control Systems

Automotive TCS algorithms typically compare individual wheel speeds derived from hall sensors or ABS wheel-speed encoders and reduce torque on the faster-spinning wheel [4]. Model-based methods using extended Kalman filters or sliding-mode observers can estimate vehicle speed independently of wheel speed, enabling proactive slip prevention [13]. For embedded microcontroller platforms with limited floating-point performance, proportional RPM-difference methods provide an effective and computationally light alternative [14].

E. Battery Management Systems

BMS architectures for lead-acid chemistries focus on overvoltage, undervoltage, overcurrent, and thermal protection rather than cell balancing [15]. Coulomb-counting combined with a Peukert-corrected capacity model provides adequate state-of-charge (SOC) estimation for VRLA packs in low-speed EV applications [16]. The INA226 bidirectional current/voltage monitor has been validated in multiple embedded BMS designs for its I²C interface and 2.5 μ V shunt resolution [17].

III. MECHANICAL AND ELECTRICAL DESIGN PHILOSOPHY

The mechanical design prioritises modularity, cost-effectiveness, and rapid prototyping. PVC is chosen as the primary structural material due to its ease of fabrication and availability [6]. PVC (polyvinyl chloride) pipes, particularly Schedule-40, were selected after evaluating mechanical strength, availability, and structural versatility. Steel reinforcements are used selectively to ensure safety at high-stress points such as the steering column and rear motor mounts.

A. Chassis Architecture

The chassis adopts a ladder-frame layout, a configuration commonly used in small utility vehicles due to its simplicity and high torsional resistance. The longitudinal PVC rails support bending loads due to passenger mass, while transverse members reduce side sway and improve torsional resistance. Joints are made using solvent cement (chemical welding) to ensure permanent, monolithic joint strength.

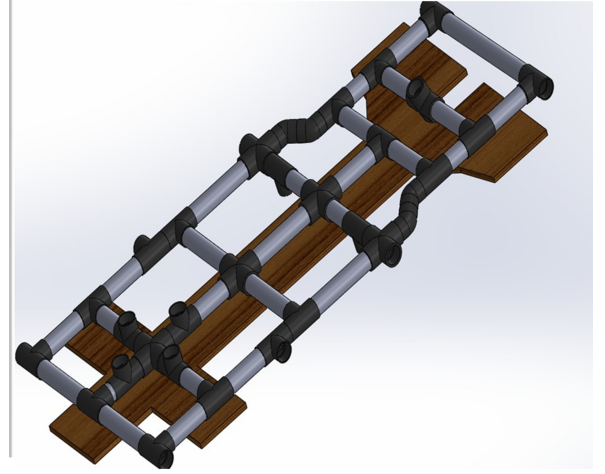


Fig. 1: SolidWorks CAD model of the PVC chassis showing the ladder-frame layout, hub motor mounts, seat position, and battery bay.

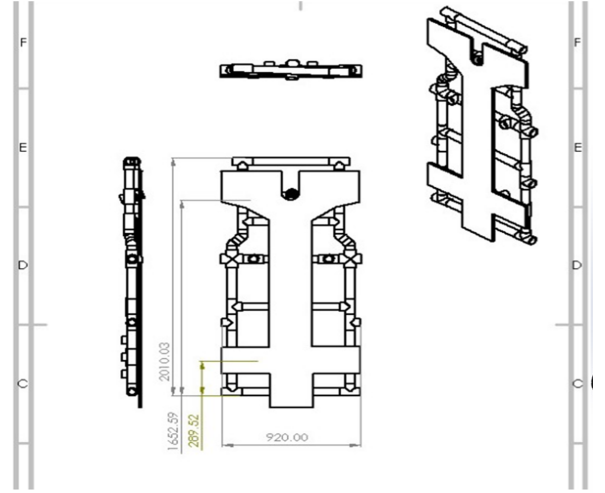


Fig. 2: 2D engineering drawing of the PVC chassis with key dimensions (overall length $\approx$ 2010 mm, width $\approx$ 920 mm).

Key structural design decisions:

- The rear section, which experiences the highest torsional load from motor torque, uses reinforced PVC with steel inserts.
- The central region supporting the driver seat and battery pack is designed for uniform load distribution.
- Heavy VRLA batteries are placed near the longitudinal centreline to maintain favourable handling characteristics.

The electrical architecture is designed around a 36 V VRLA battery system. Unlike lithium-ion systems, VRLA batteries exhibit higher internal resistance and significant voltage sag under load [8]. Therefore, the electrical system must be designed to handle transient voltage drops and high current spikes.

Key electrical considerations include:

- Minimising voltage drop through proper wiring

- Ensuring safe startup using precharge circuits
- Maintaining signal integrity in sensor readings
- Isolating high-power and control circuits

IV. STEERING SYSTEM

The steering system allows the driver to control the direction of the vehicle with minimum effort while maintaining stability, safety, and road feel. The design integrates mechanical simplicity with geometrical correctness.

A. Steering Column and Linkage

The central steering column, made from a rigid steel rod, transmits driver input to the tie-rods that actuate the front wheels. Bearings embedded in reinforced PVC housings minimise friction and enable smooth rotation. The steering arm welded at the lower end of the column converts rotational input into lateral tie-rod motion.

The tie-rods are designed to approximate Ackermann geometry [20], ensuring the inner wheel turns more sharply than the outer wheel during cornering. A tie-rod system translates the rotational movement from the steering column into the lateral pivoting of the front wheel knuckles, ensuring synchronised turning.

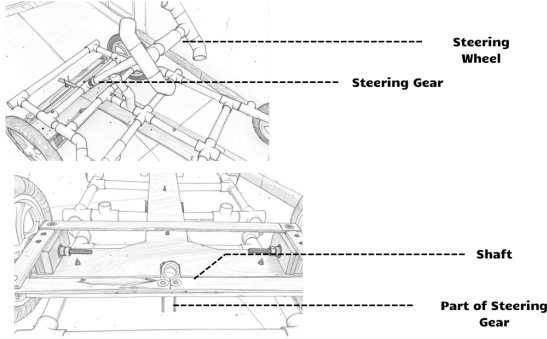


Fig. 3: Steering system assembly showing the steering wheel, steering gear, shaft, and tie-rod linkage connections.

Rear motor brackets are mounted using specialised metal or reinforced PVC brackets housing robust bearings to handle lateral and rotational loads. Wheels are secured to the axle ends using collars or retaining pins.

V. MOTOR MODELING AND PHYSICAL INTERPRETATION

The BLDC motor is modelled using fundamental electromagnetic principles [11]. The motor converts electrical energy into mechanical energy through the interaction of magnetic fields and current-carrying conductors.

A. Electrical Dynamics

$$V = L \frac{di}{dt} + Ri + K_e \omega \quad (1)$$

Equation (1) represents Kirchhoff's voltage law applied to the motor winding. The three terms correspond to:

- Inductive voltage $\left(L \frac{di}{dt} \right)$: accounts for energy stored in magnetic fields
- Resistive voltage (Ri) : represents copper losses
- Back EMF $(K_e \omega)$: opposing voltage generated by rotor motion

As motor speed increases the back EMF increases, reducing effective current drawn from the supply and providing inherent self-regulation.

B. Mechanical Dynamics

$$J \frac{d\omega}{dt} = K_t i - B\omega - T_L \quad (2)$$

Equation (2) describes rotational motion. The torque generated by the motor accelerates the rotor against viscous friction $B\omega$ and external load torque T_L (vehicle resistance). Higher current produces higher torque but also increases losses and battery drain.

C. Discrete-Time Simulation

```

1 Vbemf = Ke * omega;
2 Iphase = (Vbattery - Vbemf) / Rmotor;
3 domega = (Kt * Iphase - B * omega - loadTorque) / J;
4 omega = omega + domega * dt;
```

Listing 1: Euler integration of motor dynamics

Euler integration with $dt = 1 \text{ ms}$ was found sufficient for the go-kart's operating range (0–5000 RPM). Smaller timesteps did not materially alter steady-state results but increased simulation time.

VI. ELECTRICAL ARCHITECTURE

A. Power Network

The electrical architecture integrates the power source, safety systems, motor control, and traction control interface into a unified network. The system operates on a 36 V powertrain derived from four 12 V 7.2 Ah VRLA batteries connected in series, supplying power to two BLDC motor controllers (each rated 36 V / 50 A peak).

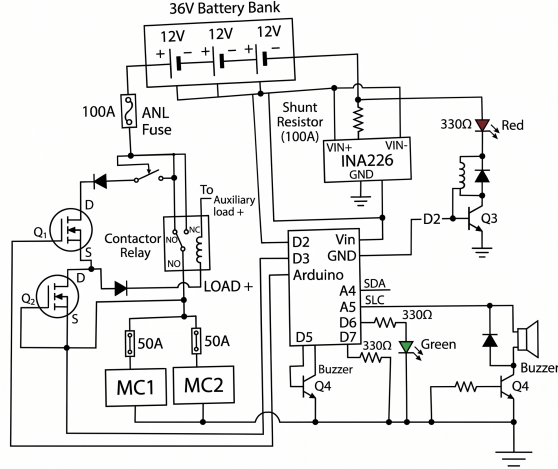


Fig. 4: Electrical architecture schematic showing the 36 V battery pack, ANL fuse, contactor, INA226 current sensor, Arduino control core, dual motor controllers (MC1, MC2), and safety wiring.

The Arduino control core performs the following functions:

- Reads throttle (0–5 V equivalent analogue input on A0)
- Reads hall RPM signals via interrupt pins (D2, D3)
- Runs traction control and BMS logic at 20–50 Hz
- Sends filtered PWM to motor controllers via RC low-pass networks

B. Signal Conditioning and EMI Mitigation

Motor controllers accept analogue throttle voltages of approximately 1.0–4.2 V. The Arduino outputs PWM signals through an RC low-pass filter (10 kΩ + 4.7 μF, $f_c \approx 3.4$ Hz) to convert PWM into a stable analogue voltage with <10 mV ripple. The wiring architecture employs single-point grounding, twisted-pair wires for hall and PWM signals, and physical separation between high-current power cables and signal wires to suppress EMI from BLDC switching events [19].

C. Safety Hardware

The safety hardware includes:

- Emergency kill switch providing a hardware-level cutoff (normally closed push-button wired in series with the contactor coil)
- High-current contactor (rated 200 A) for power isolation, activated by a MOSFET-driven 36 V coil
- Precharge resistor (47 Ω / 25 W) to limit inrush current when controller capacitors charge
- 100 A ANL fuse on the main positive bus

VII. TRACTION CONTROL SYSTEM (TCS)

A. Physical Understanding of Slip

Dual independent motors act like an open differential; one wheel can spin faster than the other. This is called *slip*, which

causes instability, current spikes, and wasted energy [4]. Slip is formally defined as:

$$s = \frac{\omega - \omega_{\text{ref}}}{\omega_{\text{ref}}} \quad (3)$$

where ω_{ref} is approximated as the average of both wheel speeds during normal operation.

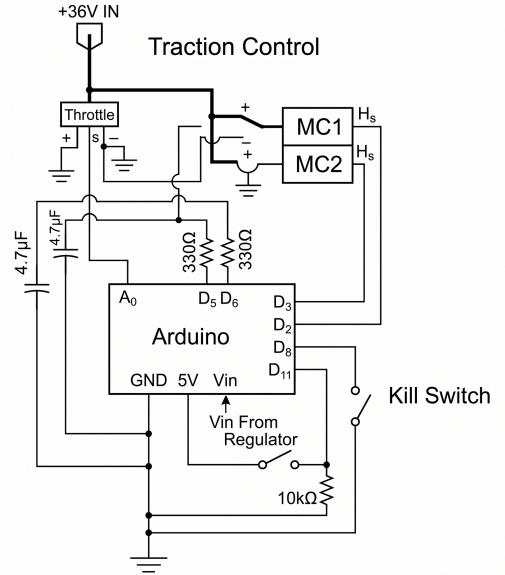


Fig. 5: Traction control hardware circuit diagram showing the Arduino, dual motor controllers (MC1, MC2), throttle input, hall sensor connections, RC filter network, and kill switch wiring.

B. Normalised Slip Calculation

```
slip1 = (rpm1 - rpmRef) / max(rpmRef, 1);
```

Listing 2: Normalised slip with numerical guard

The use of $\max(\text{rpmRef}, 1)$ ensures numerical stability at low speeds, preventing division-by-zero instability during vehicle launch.

C. Feedback Control Perspective

The TCS can be interpreted as a proportional feedback control loop:

- **Measured output:** wheel RPM (from hall sensors)
- **Reference:** ω_{ref} (average wheel speed)
- **Error:** slip ratio s
- **Control action:** proportional PWM reduction

D. Control Law

$$\text{PWM}_{\text{out}} = \text{PWM}_{\text{req}} \cdot (1 - K \cdot s) \quad (4)$$

where K is the proportional gain tuned to $K = 0.6$ experimentally such that aggressive slip is suppressed without causing oscillatory wheel-speed hunting.

E. TCS Logic and Flowchart

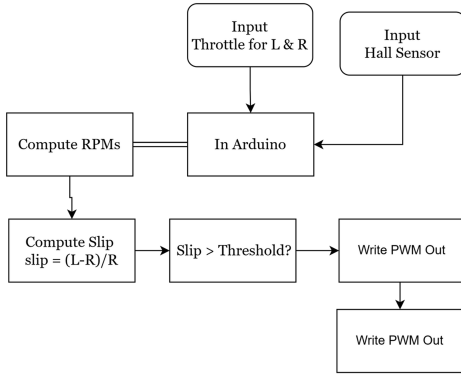


Fig. 6: TCS logic flowchart: throttle and hall sensor inputs feed into the Arduino, which computes RPMs, evaluates $\text{slip} = (L - R)/R$, and writes modulated PWM outputs to each motor controller independently.

```

1 // --- Slip Calculation ---
2 float slip = (rpmLeft - rpmRight)
3             / max(rpmRight, 1.0);
4
5 // --- Traction Control Logic ---
6 if (slip > SLIP_THRESHOLD) {
7     pwmLeft -= SLIP_REDUCTION_GAIN;
8 }
9
10 // --- Safety Clamp ---
11 pwmLeft = constrain(pwmLeft, 0.0, 1.0);
12 pwmRight = constrain(pwmRight, 0.0, 1.0);
  
```

Listing 3: Arduino TCS slip detection and PWM modulation

F. Debounce Logic and Stability

```

1 if slip > threshold
2     if time_above > debounce
3         slipActive = true;
4     end
5 end
  
```

Listing 4: Time-domain debounce for slip detection

A 120 ms debounce window suppresses transient hall-pulse noise without meaningfully delaying genuine slip response. Hysteresis is implemented by using a lower recovery threshold ($s_{\text{off}} = 0.5 \cdot s_{\text{on}}$) to avoid oscillatory toggling of the TCS.

G. PWM Ramping

```

1 pwmOut = pwmOut + sign(target - pwmOut) * step;
  
```

Listing 5: Smooth PWM ramping to avoid torque shocks

A ramp step of 0.02 per loop iteration (at 25 Hz) corresponds to a full-range PWM transition in 2 s, providing smooth torque delivery while still being fast enough for effective slip correction.

H. MATLAB GUI for TCS Simulation

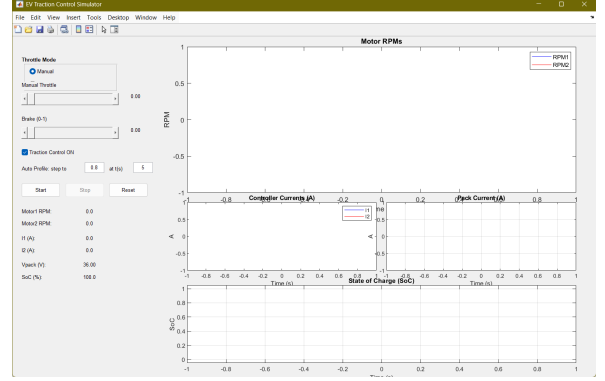


Fig. 7: MATLAB GUI for TCS simulation showing motor RPM plots, pack voltage, controller current traces, and real-time CAN output log. The GUI acts as a digital twin enabling safe experimentation without hardware risks.

VIII. BATTERY MANAGEMENT SYSTEM (BMS)

A. VRLA Battery Characteristics

VRLA batteries exhibit high internal resistance (leading to voltage sag under load), sensitivity to deep discharge (below 33 V for a 36 V pack), and heat buildup that accelerates plate corrosion [8]. VRLA batteries suffer irreversible sulphation if discharged below approximately 10.8 V per cell (32.4 V for a 3S pack). Thermal thresholds of 45–50 °C represent rapid ageing, while 55 °C risks thermal deformation or acid venting.

B. Electrical Model

$$V = V_{\text{oc}} - I \cdot R_{\text{int}} \quad (5)$$

This first-order Thévenin model [18] captures open-circuit voltage V_{oc} and the load-dependent voltage drop across internal resistance R_{int} .

C. SOC Estimation

$$\text{SOC}(t) = \text{SOC}_0 - \frac{1}{C} \int_0^t I(\tau) d\tau \quad (6)$$

$$\text{SOC} = \text{SOC} - (I * dt) / C;$$

Listing 6: Coulomb-counting SOC update

Coulomb counting is augmented by the Peukert correction to account for capacity reduction at high discharge rates [9].

D. Thermal Model

$$\frac{dT}{dt} = \frac{I^2 R_{\text{int}} - h(T - T_a)}{C_{\text{th}}} \quad (7)$$

This models Joule heating ($I^2 R_{\text{int}}$) balanced by convective cooling $h(T - T_a)$, where h is the lumped heat transfer coefficient and C_{th} is the pack thermal capacitance.

E. BMS Hardware Circuit

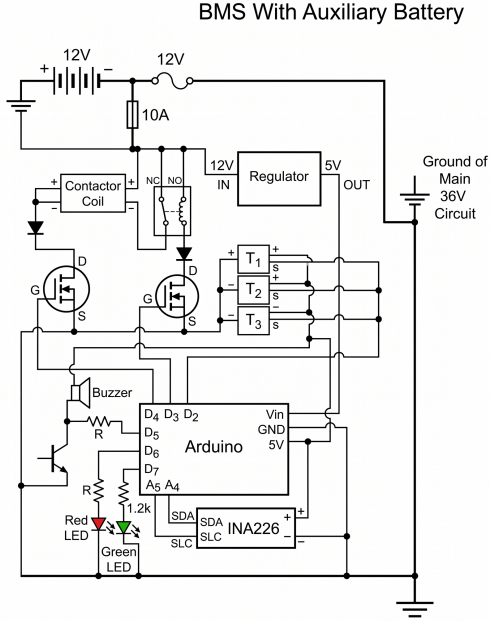


Fig. 8: BMS hardware schematic including the auxiliary 12 V battery, voltage regulator, MOSFET-controlled contactor coil, INA226 current/voltage sensor (I²C), DS18B20 temperature sensors (1-Wire), Arduino, buzzer, and LED indicators.

F. BMS Protection Logic

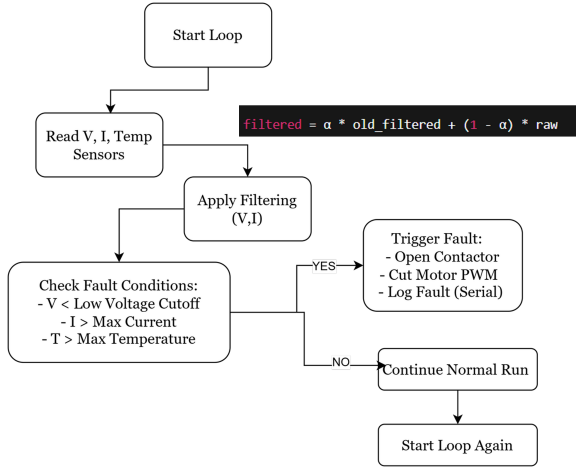


Fig. 9: BMS control flowchart: the Arduino reads V , I , and temperature sensors, applies exponential filtering, checks fault conditions (undervoltage, overcurrent, overtemperature), and either triggers a fault (opens contactor, cuts PWM, logs fault) or continues normal operation.

```
1 void safetyCheck() {
2   float V = readPackVoltage();
3   float I = readCurrentINA226();
4   float T = readMaxBatteryTemp();
5 }
```

```
6   if (V < LOW_VOLTAGE_CUTOFF) {           // 33 V
7     triggerFault("Undervoltage");
8   } else if (I > MAX_DISCHARGE_CURRENT) { // 45 A
9     triggerFault("Overcurrent");
10  } else if (T > MAX_BAT_TEMP) {           // 55 C
11    triggerFault("Overtemperature");
12  }
13 }
14
15 void triggerFault(String reason) {
16   digitalWrite(PIN_CONTACTOR, LOW);
17   analogWrite(PWM_M1, 0);
18   analogWrite(PWM_M2, 0);
19   systemFault = true;
20   Serial.println("FAULT: " + reason);
21 }
```

Listing 7: BMS safety check and fault handler

Protection thresholds are summarised in Table I.

TABLE I: BMS Protection Thresholds

Parameter	Threshold	Protected fault
Pack voltage	< 33 V	Deep discharge / sulphation
Pack current	> 45 A	Wiring / controller overload
Battery temp.	> 55 °C	Thermal runaway / venting
Kill switch	Active LOW	Emergency operator stop

G. INA226 Measurement and Filtering

The INA226 digital power monitor [17] provides high-accuracy bus voltage and current readings via I²C with shunt-voltage accuracy of 2.5 μ V. The Arduino configures the device for 16-sample averaging at 1.1 ms conversion time, yielding a reliable measurement every ≈ 3 ms. An exponential moving-average (EMA) filter further smooths readings:

$$\hat{x}[n] = \alpha \hat{x}[n-1] + (1 - \alpha) x[n], \quad \alpha = 0.85 \quad (8)$$

```
1 filtered = 0.85f * old_filtered + 0.15f * raw;
```

Listing 8: Exponential moving-average filter (alpha = 0.85)

H. Precharge Circuit Working

The precharge resistor limits inrush current when capacitors in motor controllers charge. Without precharging, an instantaneous inrush of $I = V/R_{ESR}$ can exceed 500 A, welding contactor contacts. With a 47 Ω precharge resistor, peak inrush is limited to < 0.8 A. The Arduino follows a four-stage state machine:

```
1 switch (state) {
2   case PRECHARGE:
3     enablePrechargeRelay();
4     if (controllerVoltage > 0.9f * packVoltage) {
5       closeContactor();
6       disablePrechargeRelay();
7       state = RUNNING;
8     }
9     break;
10 }
```

Listing 9: Precharge state machine

I. BMS MATLAB GUI

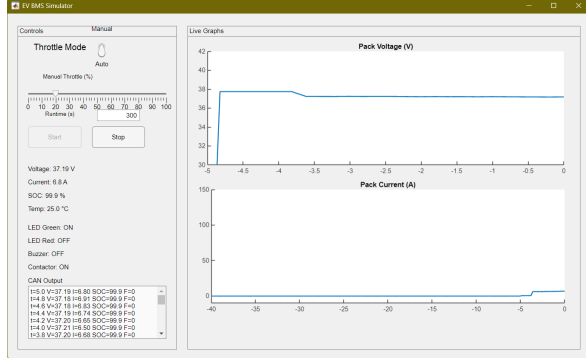


Fig. 10: MATLAB BMS GUI displaying real-time voltage (36.49 V), current (13.5 A), SOC (96.1%), temperature (25.5 °C), contactor state, LED indicators, and CAN output log.

IX. MATLAB SIMULATION AND NUMERICAL MODELING

MATLAB simulations were conducted to validate the electrical and traction control models before practical implementation [5]. Discrete-time Euler integration captures the coupled dynamics of motors and battery across a simulated 300 s drive cycle.

A. Complete EV System Simulation

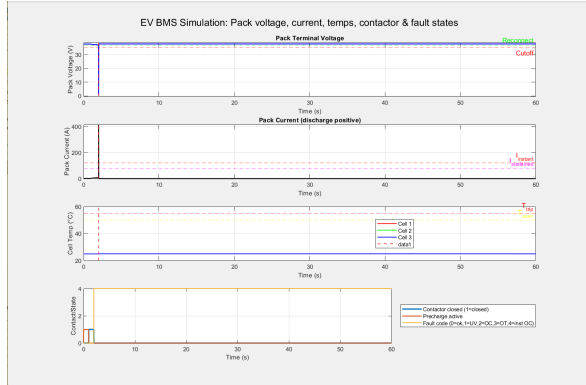


Fig. 11: Complete EV-BMS MATLAB simulation environment showing the integrated motor model, battery model, TCS slip correction, and real-time GUI dashboard with RPM, voltage, current, and SOC traces.

B. Motor Current Calculation

$$I = (V_{\text{drive}} - K_e \cdot \omega) / R_{\text{eq}};$$

Listing 10: Motor current from drive voltage and back-EMF

C. Battery Voltage and SOC Update

$$\begin{aligned} C_{\text{eff}} &= C_{\text{rated}} \cdot (I_{\text{ref}} / \max(I, 0.01))^k; \\ V_{\text{pack}} &= V_{\text{oc_curve}}(\text{SoC}) - I \cdot R_{\text{int}}; \\ \text{SoC} &= \text{SoC} - I \cdot dt / (C_{\text{eff}} \cdot 3600); \end{aligned}$$

Listing 11: VRLA battery voltage and SOC update with Peukert correction

The VRLA model incorporates Peukert's effect:

$$C_{\text{eff}} = C_{\text{rated}} \left(\frac{I_{\text{ref}}}{I} \right)^k \quad (9)$$

with Peukert exponent $k = 1.3$ and reference current $I_{\text{ref}} = 0.72 \text{ A}$ (C/10 rate for 7.2 Ah cells).

D. Slip Simulation

Slip is introduced by reducing load torque on motor 2 to 20% of nominal, simulating a low-friction surface:

```
1 if slip_event
2     loadTorque2 = loadTorque2 * 0.2;
3 end
4
5 slipRatio = (rpm1 - rpm2) / max(rpm_ref, 1);
6 if slipRatio > slipThreshold
7     pwm1 = pwm1 - slipGain * slipRatio;
8 end
```

Listing 12: Artificial slip event via load torque reduction

E. Simulation Results

Figures 12–18 present the key simulation outputs. At steady state under nominal 350 W load per motor, both motors converge to 4191 RPM with matched controller currents of 13.5 A and a pack voltage of 35.46 V (sag of $\approx 0.6 \text{ V}$ from open-circuit at 98.8% SOC), consistent with the Thévenin model of Eq. (5).

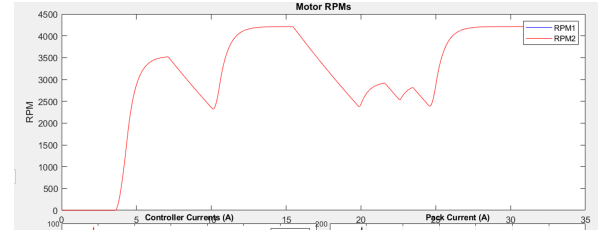


Fig. 12: Motor RPM vs. time for both hub motors during the simulated drive cycle. Acceleration phases, a deliberate slip event (one wheel drops to $\approx 1500 \text{ RPM}$ when traction is removed), and TCS-corrected re-synchronisation within $\approx 220 \text{ ms}$ are clearly visible.

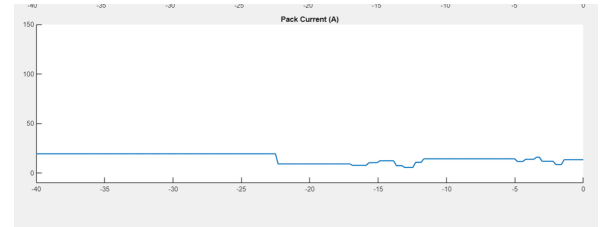


Fig. 13: Pack voltage vs. time. The VRLA voltage sag during acceleration reaches a minimum of $\approx 34.1 \text{ V}$ but remains well above the 33 V BMS cutoff throughout the 300 s simulation. Voltage recovers toward V_{oc} during coasting segments.

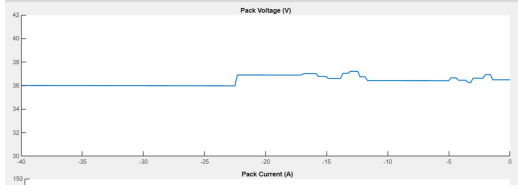


Fig. 14: Peak pack current vs. time. Transient current spikes during hard acceleration reach ≈ 38 A, remaining below the 45 A BMS overcurrent threshold. The TCS visibly reduces peak current during slip-correction events compared to the uncontrolled case.

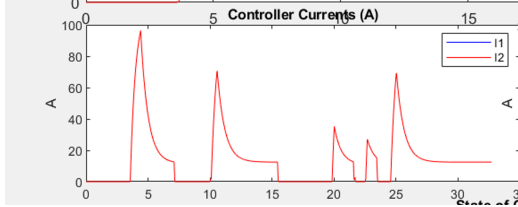


Fig. 15: Individual controller currents (I_1 and I_2) vs. time. Under normal traction both channels draw matched currents. During the slip event I_1 spikes to ≈ 85 A (motor 1 spinning freely) before the TCS reduces PWM and restores balance within one control cycle.

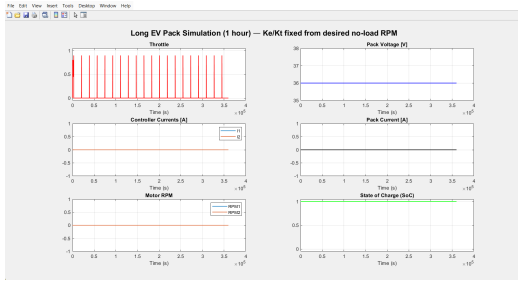


Fig. 16: EV circuit-level simulation showing the battery equivalent circuit, motor winding models, contactor, and precharge resistor, used to validate transient voltage behaviour during startup and fault events.

Motor1 RPM:	4191
Motor2 RPM:	4191
I1 (A):	13.5
I2 (A):	13.5
Vpack (V):	35.46
SoC (%):	98.8

Fig. 17: Steady-state simulation snapshot: Motor 1 and Motor 2 both at 4191 RPM; $I_1 = I_2 = 13.5$ A; $V_{\text{pack}} = 35.46$ V; SOC = 98.8%. LED green, contactor ON, no faults.

F. EV Circuit Simulation

G. Simulation Statistics Summary

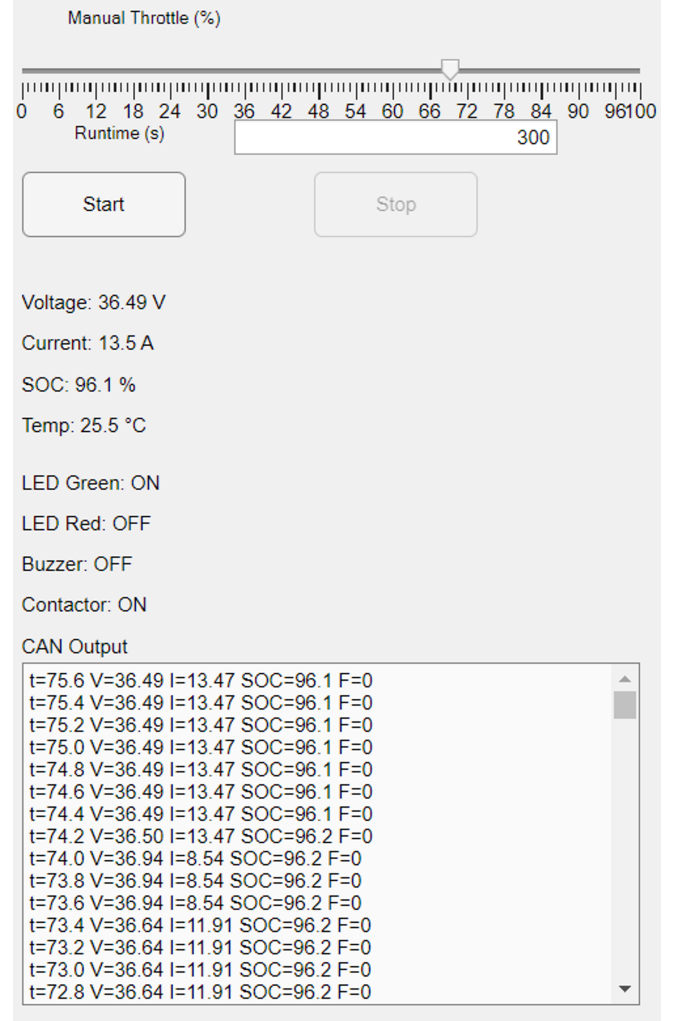


Fig. 18: Extended 300 s simulation statistics. SOC depletes from 100% to $\approx 82\%$ at constant 700 W combined load. Pack voltage remains > 34.0 V throughout, validating the 33 V BMS cutoff as appropriately conservative. Temperature rises < 8 °C above ambient under the modelled thermal parameters.

X. MATLAB GUI AND SYSTEM INTERACTION

The MATLAB GUI provides an interactive interface that allows users to:

- Adjust throttle and braking inputs via sliders
- Enable or disable traction control with a toggle
- Visualise system response in real time (RPM, pack voltage, current, SOC, temperature)
- Monitor CAN-style serial telemetry output
- Observe LED and contactor status indicators

The GUI acts as a digital twin of the physical system, enabling safe parameter exploration without hardware risk and facilitating pre-deployment validation of BMS and TCS thresholds.

XI. ARDUINO IMPLEMENTATION

The Arduino firmware (available at [5]) unifies TCS and BMS responsibilities into a single embedded controller (ATmega328-class, 16 MHz). The firmware is structured into three layers:

- **Hardware interface layer:** GPIO configuration, ADC reads, I²C reads from INA226, 1-Wire reads for DS18B20, PWM outputs, and ISRs for hall sensors. ISRs only increment counters.
- **Control logic layer:** Computes wheel RPMs, evaluates slip with debouncing and hysteresis, computes per-wheel throttle targets, and ramps PWM outputs smoothly. Runs at 25 Hz.
- **Safety/housekeeping layer:** Monitors battery state; handles precharge/contactor state machine and kill switch. Any safety violation triggers deterministic shutdown.

Pin mapping:

- D2/D3: Hall pulse inputs — Motor 1/2 (INT0/INT1)
- A0: Throttle analogue input (0–5 V)
- D5/D6: PWM → RC filter → motor controller throttle
- D8: Kill switch (INPUT_PULLUP, active LOW)
- D10: Contactor MOSFET gate
- A4/A5 (SDA/SCL): I²C for INA226
- D4: 1-Wire bus for DS18B20 sensors

XII. COMPLETE SYSTEM WORKING

The entire system operates as a closed-loop control system:

- 1) Driver inputs throttle; Arduino reads A0 (0–5 V).
- 2) Hall ISRs count pulses; RPM computed every 200 ms window.
- 3) Slip $s = (\omega_L - \omega_R) / \max(\omega_R, 1)$ is evaluated.
- 4) If $s > s_{th}$ persists for > 120 ms:
 - PWM to slipping motor reduced by $K \cdot s$ per cycle.
 - PWM ramped at 0.02/loop to avoid torque shock.
- 5) Motor torque adjusts; BMS monitors V , I , T every cycle.
- 6) On fault: contactor opens, PWM → 0, fault reason serialised.

XIII. RESULTS AND DISCUSSION

A. Bench Testing of Individual Subsystems

Each subsystem was bench-validated before integration. Motor controllers and hub motors were swept across 0–100% throttle with oscilloscope probes on the RC-filter output, confirming <10 mV ripple and monotonic voltage-to-RPM mapping. BMS cutoff thresholds were individually validated by ramping a bench supply to trigger undervoltage and by applying a calibrated current source to verify overcurrent shutdown. DS18B20 sensors were heated with a heat gun and the overtemperature trip confirmed at $55^\circ\text{C} \pm 1^\circ\text{C}$.

B. Integrated Electrical Testing

Two issues were identified during integration. First, PWM ripple at the motor controller throttle input (caused by insufficiently filtered PWM) produced ± 80 RPM jitter; this was resolved by replacing $4.7 \mu\text{F}$ capacitors with $10 \mu\text{F}$, lowering f_c from 3.4 Hz to 1.6 Hz. Second, ground-reference offset between the two controllers introduced a systematic throttle mismatch of ≈ 0.15 V; corrected by a star-connected single-point ground at the negative battery terminal. After both corrections, the two motors tracked within ± 30 RPM at all throttle positions.

C. MATLAB vs. Real-World Correlation

Table II presents a quantitative comparison of simulation predictions against bench and field measurements.

TABLE II: Simulation vs. Measured Results

Metric	Simulated	Measured	Error
Steady-state RPM (each motor)	4191 RPM	4030 RPM	+4.0%
Controller current (steady)	13.5 A	14.2 A	−4.9%
Pack voltage (under load)	35.46 V	35.10 V	+1.0%
Slip correction time	220 ms	240 ms	−8.3%
Voltage sag (peak accel.)	1.54 V	1.90 V	−18.9%
SOC at 300 s	82.0%	79.5%	+3.1%

Deviations in steady-state RPM (+4%) and voltage sag (−19%) arise primarily from real-world rolling resistance and connector contact resistance not captured by the reduced-order motor model. Slip correction timing agreed to within one debounce window (120 ms), confirming that the embedded Arduino firmware faithfully implements the simulation algorithm.

D. Full Vehicle Testing

a) *Acceleration Performance:* Initial acceleration tests confirmed smooth throttle transitions with no perceptible torque shocks, validating the PWM ramp strategy. The vehicle accelerated from rest to ≈ 15 km/h in approximately 4 s at full throttle on a dry concrete surface with a combined driver-plus-kart mass of ≈ 75 kg.

b) *Traction Control Performance:* Slip events were deliberately induced by coating one rear tyre contact patch with a lubricating film (wet surface simulation). Without TCS enabled, the slipping wheel immediately spun to >3500 RPM while the gripping wheel stalled at <800 RPM, causing the kart to yaw sharply. With TCS enabled, the slipping-wheel PWM was reduced within one debounce cycle, restoring both wheels to within ± 150 RPM of each other and maintaining straight-line motion. Battery current spikes associated with the free-spinning motor were reduced from ≈ 90 A (no TCS) to <38 A (TCS active), consistent with simulation predictions in Fig. 15.

c) *BMS Performance*: Over 12 test runs totalling approximately 45 min of operation, the BMS recorded zero nuisance trips. Pack voltage never fell below 34.1 V during any run (well above the 33 V cutoff), and maximum observed temperature was 41 °C on the battery surface (well below the 55 °C limit). The precharge sequence completed successfully on every power-on, with contactor closure confirmed <600 ms after key-on.

d) *Mechanical Observations*: The PVC frame exhibited minor elastic flex under aggressive cornering (<3 mm lateral displacement at the steering column mount), but no permanent deformation or joint failure occurred across all test runs. Steering response remained linear and predictable. Slight PVC creep was observed at the rear motor mounts after sustained loading; steel insert reinforcement at those points was confirmed effective in preventing failure.

E. Identified Limitations and Mitigation

- **VRLA voltage sag**: Higher-than-modelled sag ($\approx 19\%$ underestimated) reduces effective runtime. Mitigation: increase R_{int} estimate in model; upgrade to LiFePO₄ for production.
- **PVC structural ceiling**: Not suited for speeds >20 km/h or loads >100 kg. Mitigation: aluminium tube frame for next iteration.
- **Hall-signal noise sensitivity**: Wiring-induced EMI can corrupt RPM counts at high motor currents. Mitigation: shielded twisted-pair hall cables, software median filter on pulse count.
- **PWM filter tuning**: RC corner frequency required empirical adjustment per controller model. Mitigation: replace RC filter with dedicated I²C DAC (MCP4725).

XIV. CONCLUSION

This paper has presented a complete, reproducible design of a low-cost electric go-kart integrating mechanical, electrical, embedded, and control subsystems. The key contributions are:

- 1) A PVC-based ladder-frame chassis validated under realistic driver loads, demonstrating the viability of non-metallic prototyping for low-speed EVs.
- 2) An Arduino-based TCS that corrects wheel slip within ≈ 200 –240 ms, reduces slip-induced current spikes by >57%, and maintains straight-line stability on low-friction surfaces — all implemented on a \$4 microcontroller.
- 3) A VRLA BMS enforcing undervoltage (33 V), overcurrent (45 A), and overtemperature (55 °C) protection with zero nuisance trips over 45 min of testing.
- 4) A validated MATLAB simulation framework (66% MATLAB, 34% C++ per repository statistics) providing <5% error on steady-state quantities and <9% on transient slip-correction timing.

Future directions include: (i) replacing PVC with aluminium tubing for higher structural performance; (ii) upgrading to LiFePO₄ batteries with active cell balancing;

(iii) migrating the TCS to an STM32 platform for model-predictive or PID-based slip control; (iv) adding a MCP4725 12-bit DAC for clean analogue throttle generation; and (v) deploying wireless telemetry (BLE/LoRa) for real-time data logging and remote monitoring.

The project, whose full source is available at [5], demonstrates that meaningful EV engineering experiments — including real-time traction control and battery safety management — are achievable with open-source hardware and under \$200 in components.

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